



Statistics of Financial Markets

Chapter 8: Stochastic Volatility Models – Part 2



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Outline

Motivation

EGARCH Model

PGARCH Model

TGARCH Model

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Comparison

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References

Motivation

- Limitations of symmetric GARCH: inability to capture the leverage effect
- Asymmetric response of volatility to positive vs. negative shocks
- Need for flexible volatility specifications in risk management

EGARCH Model

- Nelson's (1991) Exponential GARCH specification
- Log-volatility equation: $\ln \sigma_t^2 = \omega + \beta \ln \sigma_{t-1}^2 + \gamma z_{t-1} + \alpha(|z_{t-1}| - \mathbb{E}|z_{t-1}|)$
- No positivity constraints required
- Captures asymmetry through the sign parameter γ

PGARCH Model

- Ding, Granger, and Engle's (1993) Power GARCH
- $\sigma_t^\delta = \omega + \sum_{i=1}^q \alpha_i (|\varepsilon_{t-i}| - \gamma_i \varepsilon_{t-i})^\delta + \sum_{j=1}^p \beta_j \sigma_{t-j}^\delta$
- Flexible power parameter δ nesting GARCH and AVGARCH
- Asymmetry through leverage parameters γ_i

TGARCH Model

- Zakoian's (1994) Threshold GARCH specification
- $\sigma_t = \omega + \sum_{i=1}^q (\alpha_i \varepsilon_{t-i}^+ - \gamma_i \varepsilon_{t-i}^-) + \sum_{j=1}^p \beta_j \sigma_{t-j}$
- GJR-GARCH as a related threshold model (Glosten, Jagannathan, Runkle, 1993)
- Piecewise linear response to shocks

GARCH-M Model

- GARCH-in-Mean: risk premium in the conditional mean equation
- $r_t = \mu + \lambda g(\sigma_t^2) + \varepsilon_t$
- Common specifications: $g(\sigma_t^2) = \sigma_t^2$, $g(\sigma_t^2) = \sigma_t$, $g(\sigma_t^2) = \ln \sigma_t^2$
- Testing the risk-return tradeoff

Comparison

- ▣ News impact curves across different GARCH specifications
- ▣ Nesting relationships among models
- ▣ Model selection criteria and diagnostic tests
- ▣ Strengths and weaknesses of each specification

Applications

- ▣ Fitting asymmetric GARCH models to equity returns
- ▣ Comparing forecasts from symmetric vs. asymmetric models
- ▣ Python implementation using `arch` package

 [SFM_ch8_advanced_garch](#)

Conclusions

- ▣ Summary of asymmetric and extended GARCH models
- ▣ Practical guidelines for model selection
- ▣ Connection to VaR estimation (Chapter 9)

References

- Nelson, D.B. (1991). Conditional heteroskedasticity in asset returns: A new approach.
- Ding, Z., Granger, C.W.J., Engle, R.F. (1993). A long memory property of stock market returns and a new model.
- Glosten, L.R., Jagannathan, R., Runkle, D.E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks.